

Assignment 2 Addendum: Questions about Assignments 1 and 2

Section 1: GCP & BigQuery Basics

1. Google Cloud Platform (GCP) provides cloud computing services across different models: SaaS, PaaS, and IaaS. Define each model and give an example of a GCP service that fits each category.
2. BigQuery is a cloud-based data warehouse. How does it differ from traditional on-premises databases, and what are its advantages?
3. Why is it necessary to enable the BigQuery API before using it?
4. What is a service account, and why was it required for this assignment? How does a service account differ from a user account?
5. How did you authenticate your Python script with BigQuery?

Section 2: Setting Up BigQuery & Data Upload

6. What steps did you follow to create a BigQuery project, dataset, and table?
7. Describe the structure of the database you uploaded (tables, columns, data types).
8. What Python libraries did you use to connect to BigQuery, and what was their role?
9. What steps were involved in transferring data from SQLite to BigQuery using Python?
10. What challenges did you face when uploading data, and how did you resolve them?
11. If you had to repeat this assignment for a much larger dataset, what changes would you make to your data pipeline?

Section 3: Querying Data in BigQuery

12. Write an SQL query to count the total number of rows in your BigQuery table.
13. How would you retrieve only the first 10 rows from your table in BigQuery?
14. How would you filter data for a specific date range using SQL in BigQuery?
15. What is the purpose of using LIMIT in a query, and when would you use it?
16. What does it mean to "optimize" an SQL query? Why might some queries take longer to run or cost more in BigQuery than others?

Section 4: Exploring, Validating, and Understanding Data

17. How did you use Looker to explore and visualize your BigQuery data?

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18. Did the data seem reasonable based on your analysis? What patterns or trends did you notice?
19. How did you validate your data? Did you compare it with any external sources to confirm its accuracy? If so, which sources?
20. In Assignment 1, you retrieved the data from the API and stored it in SQLite before uploading it to BigQuery in Assignment 2. Could you have skipped SQLite and loaded the data directly into BigQuery? What would be the advantages and disadvantages of doing so?